# Run report: cen-q7-cone-run-20261024T2241Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part CENSUS (sampled) under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations ($F_p$ multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

## env.txt

```
run: cen-q7-cone 20261024T2241Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 26415a19d8d07de211c7992a6c97ca99a2a3a358fabb5c02e7893639121efd5b
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/cen-q7-cone-run-20261024T2241Z CENSUS 7 cone 30000000 202610247 0 2000
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 116
```

## tool_log.txt

```
cone q=7: samples=30000000 (random) res_shaped=34702 allowed_trace_candidates=949 smooth_prime_order_hits=949 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-sample 2000)=1516/2000 secs=115.2
cone q=7: total prime-order mass estimate 217.186 +- 7.050 (hit rate per sampled form 3.163e-5)
```

## summary.txt

```
cone q=7: samples=30000000 (random) res_shaped=34702 allowed_trace_candidates=949 smooth_prime_order_hits=949 smooth_fraction_among_prime_cands=1.0000 smooth_fraction_other_res_shaped(sub-sample 2000)=1516/2000 secs=115.2
cone q=7: total prime-order mass estimate 217.186 +- 7.050 (hit rate per sampled form 3.163e-5)
```

## cand_t_hist.csv (first 12 of 177 rows)

| type | t | candidates |
|---|---|---|
| cone | −98 | 84 |
| cone | −95 | 8 |
| cone | −94 | 48 |
| cone | −93 | 7 |
| cone | −91 | 15 |
| cone | −90 | 53 |
| cone | −89 | 44 |
| cone | −88 | 44 |
| cone | −87 | 18 |
| cone | −86 | 110 |
| cone | −84 | 52 |
| cone | −83 | 52 |

## hits.csv (first 12 of 949 rows)

| type | code | t | nE | smooth |
|---|---|---|---|---|
| cone | 1623141785906 | 13 | 2389 | true |
| cone | 1935115740542 | 25 | 2377 | true |
| cone | 1654331499937 | 19 | 2383 | true |
| cone | 2676523467197 | −9 | 2411 | true |

| | | | | |
|---|---|---|---|---|
| cone | 3202755716986 | −45 | 2447 | true |
| cone | 391616815344 | −45 | 2447 | true |
| cone | 2083491091311 | 31 | 2371 | true |
| cone | 2875489973986 | 25 | 2377 | true |
| cone | 1785212176126 | −15 | 2417 | true |
| cone | 2433100403872 | −45 | 2447 | true |
| cone | 4326092431570 | 9 | 2393 | true |
| cone | 1182928668445 | 3 | 2399 | true |

## mass_by_t.csv (first 12 of 19 rows)

| type | t | nE | smooth_hits | mass_estimate | mass_stderr |
|---|---|---|---|---|---|
| cone | −75 | 2477 | 66 | 15.1046 | 1.8592 |
| cone | −71 | 2473 | 42 | 9.6120 | 1.4832 |
| cone | −65 | 2467 | 20 | 4.5771 | 1.0235 |
| cone | −57 | 2459 | 68 | 15.5623 | 1.8872 |
| cone | −45 | 2447 | 86 | 19.6817 | 2.1223 |
| cone | −39 | 2441 | 27 | 6.1791 | 1.1892 |
| cone | −15 | 2417 | 76 | 17.3932 | 1.9951 |
| cone | −9 | 2411 | 38 | 8.6966 | 1.4108 |
| cone | 3 | 2399 | 78 | 17.8509 | 2.0212 |
| cone | 9 | 2393 | 71 | 16.2489 | 1.9284 |
| cone | 13 | 2389 | 78 | 17.8509 | 2.0212 |
| cone | 19 | 2383 | 103 | 23.5723 | 2.3226 |

**Graph**

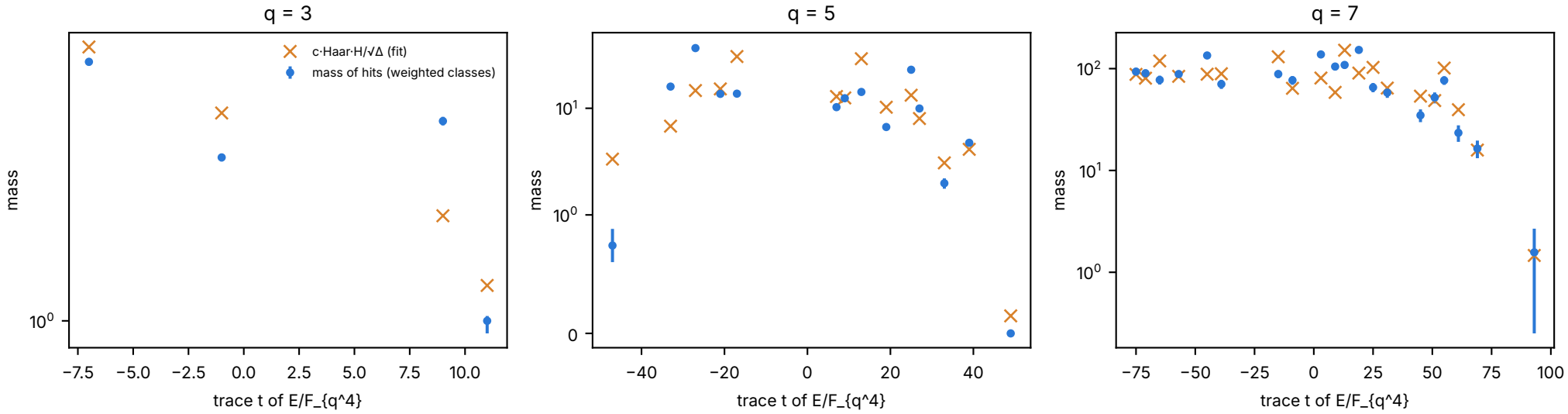



Figure 1: Mass of hits per prime-order trace with the Haar-density fit.